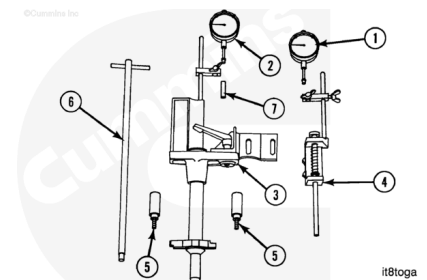


Trainings ▾

Setup

Use the injection timing tool, Part Number 3824942, or equivalent. The indicators (1 and 2) are identical.

1. Push rod travel indicator
2. Piston travel indicator
3. Piston plunger support assembly
4. Push rod plunger support assembly
5. Hold-down adapter
6. Extension assembly (adapter wrench)
7. Indicator stem extension.



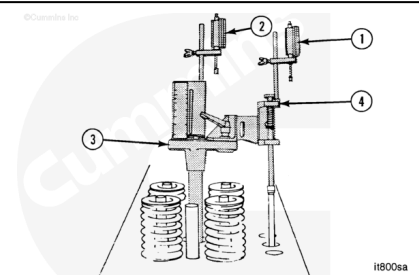
The pushrod plunger support assembly alignment is critical.

Install the pushrod plunger support (4) in the outside slot of the piston plunger support (3).

Align the pushrod plunger support with the mark. Tighten the capscrew.

Install the indicators (1 and 2) on the posts. Turn the indicators so they are **not** over the plungers.

Install the stem extension on the piston travel indicator.

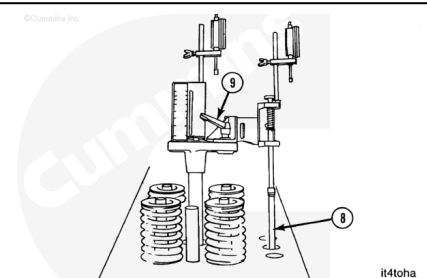


Install the injector pushrod (8) for cylinder number 1 on the right bank.

Install the timing tool in the injector bore of cylinder number 1 on the right bank. Install the hold-down adapters.

Make sure the pushrod plunger and the pushrod are aligned.

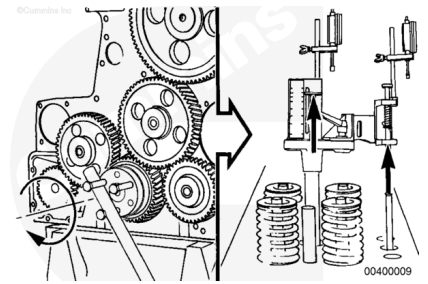
Tighten the support lock (9).



Measure

⚠ CAUTION ⚠

Only use the crankshaft to rotate the engine. If any other gear is used, a false measurement will result. Gear lash must be closed up in the direction of normal rotation.



For detailed information on the timing marks. Refer to Procedure 001-008 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-008-tr.html)

Three guide bolts equally spaced in front of the crankshaft will help rotate the engine.

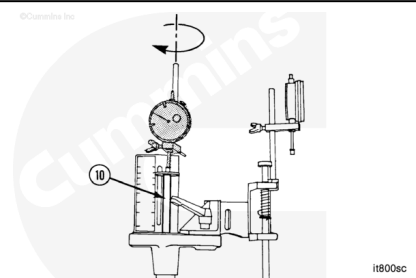
Make sure all of the gear index marks are aligned.

Turn the crankshaft in the direction of normal rotation while observing both of the timing tool plungers. Both plungers will begin moving upward when the cylinder is on the compression stroke.

Assuming all the gear index marks were aligned when the injection timing process was started, the crankshaft will have to be rotated approximately $\frac{3}{4}$ of a revolution to get to the compression stroke for cylinder number 1 on the right bank.

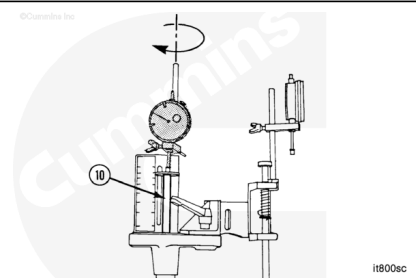
If both plungers are **not** moving up (one up and one down), the engine is on the exhaust stroke. Rotate the crankshaft one revolution to get on the compression stroke.

Establish top dead center by slowly rotating the crankshaft in the direction of normal rotation while observing the piston plunger (10). The plunger will move upward, stop, then begin to move downward. The stop point of the plunger is top dead center. Rotate the engine opposite the direction of normal rotation until the plunger begins to move down. The cylinder is now slightly before top dead center.



Turn the indicator so that the stem is touching the plunger. Carefully move the indicator downward until the needle has turned a minimum of five revolutions 12.7 mm [0.500 in]. Lock the indicator in position.

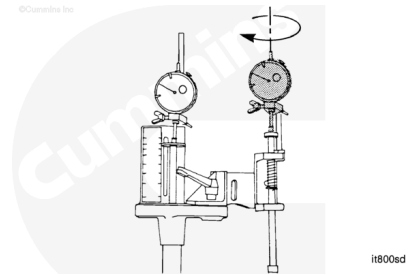
Slowly turn the crankshaft in the direction of normal rotation until the indicator needle stops turning **clockwise** (top dead center). Move the indicator down until there is **only one** revolution, 2.54 mm [0.100 in], of travel remaining until the indicator bottoms out.



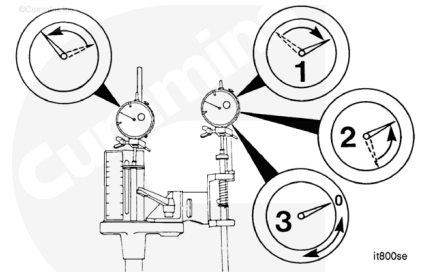
Adjust the indicator to zero.

Turn the pushrod indicator so that the stem touches the plunger.

Carefully lower the indicator until it bottoms out. Raise the indicator until the needle has turned a minimum of three revolutions, 7.62 mm [0.300 in].



Slowly turn the crankshaft in the direction of normal rotation until the pushrod indicator stops (1), momentarily reverses direction (2) (this is the crush nose on the camshaft), and stops again (3). The cam follower is now on the outer base circle of the camshaft. The piston is now approximately 45 degrees after top dead center.



It is important to record the amount of travel remaining in the pushrod travel indicator for later reference.

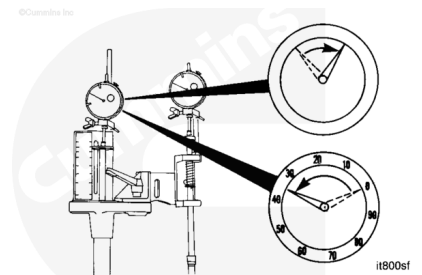
Carefully lower the push rod travel indicator until it bottoms out. Raise the indicator approximately $\frac{1}{2}$ of a revolution 1.27 mm [0.050 in]. Lock the indicator in position.

Set the indicator at zero.

Set the piston at 5.16 mm [0.203 in] before top dead center.

Observe the piston travel indicator as the crankshaft is slowly rotated in the opposite direction of normal rotation.

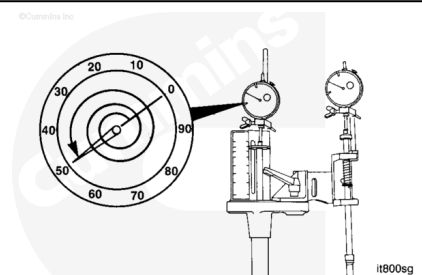
Stop rotating the crankshaft when the piston travel indicator indicates the piston is at top dead center (zero). Recheck top dead center (zero)



The crankshaft **must** be turned slowly to count the indicator revolutions accurately.

Turn the crankshaft opposite the direction of normal rotation until the indicator needle moves 2- $\frac{1}{2}$ revolutions or 6.35 mm [0.250 in].

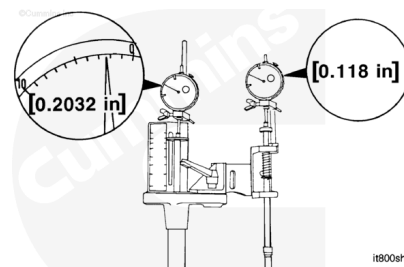
The piston is now 6.35 mm [0.250 in] before top dead center.



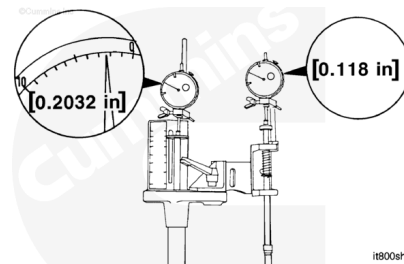
Only move the piston to 5.161 mm [0.203 in] before top dead center by turning the crankshaft in the direction of normal rotation. If the crankshaft is turned too far, it **must** be turned

the opposite direction of normal rotation more than 5.161 mm [0.203 in] before top dead center. Then, very slowly turn the crankshaft in the direction of normal rotation until the indicator indicates that the piston is 5.161 mm [0.203 in] before top dead center.

All QSK45 and QSK60 injection timing specifications are more than one indicator revolution, 2.54 mm [0.100 in].

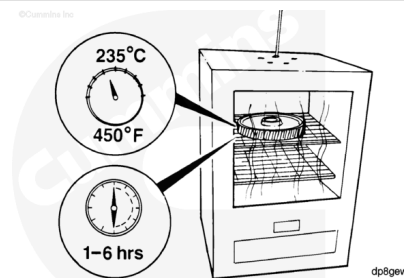


Read the pushrod travel indicator **counterclockwise** from zero. This is the injection timing measurement to compare to the specification. An example of 3 mm [0.118 in] is shown.



If unsure of the number of pushrod indicator revolutions, check by carefully lifting the indicator stem until the indicator has bottomed out. Lower the stem the amount of excess travel that was set in the third step of this procedure. Lower the stem to the plunger. Read the indicator.

If the injection timing is within specification, and a slip-fit gear is used, install the standard gear. Refer to Procedure 001-012 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-012.html>). Repeat the injection timing procedure after the camshaft gear has cooled.



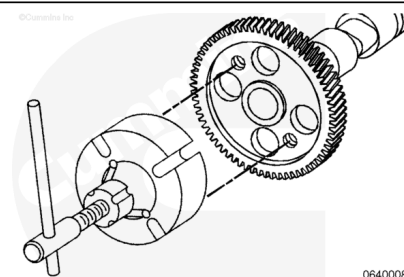
⚠ CAUTION ⚠

To avoid engine damage, the timing measurement must be confirmed after changing the key.

If the injection timing is **not** within specification, repeat the measurement procedure to check the tool setup and the zero settings.

If the timing is still **not** within specification, the camshaft key **must** be changed. For instructions to remove the camshaft gear. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-012.html>)

Record the orientation of any offset of the key. Use the following worksheet to determine an alternate key.



Repeat this procedure on a front and rear cylinder of the bank(s) from which the camshaft or camshaft key was replaced.

Table C

S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	Part Number	Direction of Offset
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite
0.235	0.033	0.039	0.0345	0.0305	0.0268	0.042	0.0252	0.029	0.037	0.0415	0.045	0.0471	0.0491	3000494	Opposite
0.215	0.031	0.037	0.0325	0.0285	0.0248	0.040	0.0232	0.027	0.035	0.0395	0.043	0.0441	0.0471	3000493	Opposite
0.0185	0.028	0.034	0.0295	0.0255	0.0218	0.037	0.0202	0.024	0.032	0.0365	0.040	0.042	0.0441	200714	Opposite
0.018	0.0275	0.0335	0.029	0.025	0.0213	0.0365	0.0197	0.0235	0.0315	0.036	0.0395	0.0415	0.0436	3000492	Opposite
0.0155	0.025	0.031	0.0265	0.0225	0.0188	0.034	0.0172	0.021	0.029	0.0335	0.037	0.039	0.0411	200706	Opposite
0.0135	0.023	0.029	0.0245	0.0205	0.0168	0.032	0.0152	0.019	0.027	0.0315	0.035	0.037	0.0391	3000491	Opposite

Table C															
0.0 11	0.0 20 5	0.0 26 5	0.0 22	0.0 18	0.0 14 3	0.0 29 5	0.0 12 7	0.0 16 5	0.0 24 5	0.0 29	0.0 32 5	0.0 34 5	0.0 06 6	20 07 08	Op po sit e
0.0 09 5	0.0 19	0.0 25	0.0 20 5	0.0 16 5	0.0 12 8	0.0 28	0.0 11 2	0.0 15	0.0 23	0.0 27 5	0.0 31	0.0 33	0.0 35 1	20 07 04	Op po sit e
0.0 07	0.0 16 5	0.0 22 5	0.0 18	0.0 14	0.0 10 3	0.0 25 5	0.0 08 7	0.0 12 5	0.0 20 5	0.0 25	0.0 28 5	0.0 30 5	0.0 32 6	20 07 09	Op po sit e
0.0 05 5	0.0 15	0.0 21	0.0 16 5	0.0 12 5	0.0 09	0.0 24	0.0 07 2	0.0 11	0.0 19	0.0 23 5	0.0 27	0.0 29	0.0 31 1	21 67 82	Op po sit e
0.0 03 3	0.0 12 8	0.0 18 8	0.0 14 3	0.0 10 3	0.0 06	0.0 21 8	0.0 05	0.0 09	0.0 16 8	0.0 21 3	0.0 24 8	0.0 26 8	0.0 28 9	20 07 11	Op po sit e
0.0 01 7	0.0 11 2	0.0 17 2	0.0 12 7	0.0 08 7	0.0 05 1	0.0 20 2	0.0 03 4	0.0 02	0.0 15 2	0.0 19 7	0.0 23 2	0.0 25 2	0.0 27 3	21 62 94	Op po sit e
0.0 00	0.0 09 5	0.0 15 5	0.0 11	0.0 07	0.0 03	0.0 18 5	0.0 01 7	0.0 06	0.0 13 6	0.0 18	0.0 20 5	0.0 23 5	0.0 25 6	S- 30 2	N/ A
0.0 01 7	0.0 11 2	0.0 17 2	0.0 09 3	0.0 05 3	0.0 01 6	0.0 16 8	0.0 00	0.0 03 8	0.0 11 8	0.0 16 3	0.0 19 8	0.0 21 8	0.0 23 9	21 62 94	Sa me
0.0 03 3	0.0 06 2	0.0 12 2	0.0 07 7	0.0 03 7	0.0 0	0.0 01 52	0.0 01 6	0.0 02	0.0 10 2	0.0 14 7	0.0 18 2	0.0 20 2	0.0 22 3	20 07 11	Sa me
0.0 05 5	0.0 04	0.0 10	0.0 05 5	0.0 01 5	0.0 02	0.0 13	0.0 03 8	0.0 00	0.0 08	0.0 12 5	0.0 16	0.0 18	0.0 20 1	21 67 82	Sa me
0.0 07	0.0 02 5	0.0 08 5	0.0 04	0.0 0	0.0 03 7	0.0 11 5	0.0 05 3	0.0 01 5	0.0 06 5	0.0 11	0.0 14 5	0.0 16 5	0.0 18 6	20 07 09	Sa me
0.0 09 5	0.0 0	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me

Table C															
0.0 11	0.0 01 5	0.0 04 5	0.0 00	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Sa me
0.0 09 5	0.0 00	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me
0.0 11	0.0 01 5	0.0 04 5	0.0 0	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Sa me
0.0 13 5	0.0 04	0.0 02	0.0 02 5	0.0 06 5	0.0 10 2	0.0 05	0.0 11 8	0.0 08	0.0 0	0.0 04 5	0.0 08	0.0 10	0.0 12 1	30 00 49 1	Sa me
0.0 15 5	0.0 06	0.0 0	0.0 04 5	0.0 08 5	0.0 12	0.0 03	0.0 13 8	0.0 10	0.0 02	0.0 02 5	0.0 06	0.0 08	0.0 10 1	20 07 06	Sa me
0.0 18	0.0 08 5	0.0 02 5	0.0 07	0.0 11	0.0 14 7	0.0 00 5	0.0 16 3	0.0 12 5	0.0 04 5	0.0 0	0.0 03 5	0.0 05 5	0.0 07 6	30 00 49 2	Sa me
0.0 18 5	0.0 09	0.0 03	0.0 07 5	0.0 11 5	0.0 15 2	0.0 0	0.0 16 8	0.0 13	0.0 05	0.0 00 5	0.0 03	0.0 05	0.0 07 1	20 07 14	Sa me
0.0 21 5	0.0 12	0.0 06	0.1 05	0.0 14 5	0.0 18 2	0.0 03	0.0 19 8	0.0 16	0.0 08	0.0 03 5	0.0 00	0.0 02	0.0 04 1	30 00 49 3	Sa me
0.0 23 5	0.0 14	0.0 08	0.0 12 5	0.0 16 5	0.2 02	0.0 05	0.0 21 8	0.0 18	0.0 10	0.0 05 5	0.0 02	0.0 00	0.0 02 1	30 00 49 4	Sa me
0.0 25 6	0.0 16 1	0.0 10	0.0 14 6	0.0 18 6	0.0 22 3	0.0 07 1	0.0 23 9	0.0 02 0	0.0 12 1	0.0 07 6	0.0 04 1	0.0 02 1	0.0 00	30 00 49 5	Sa me

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.

A working example is attached for your review to illustrate the use of this worksheet.

1.	What is the current timing?	_____
2.	What is the timing code?	_____

3.	What is the timing specification for this code ($\pm$ 0.002 in)?	_____
4.	Is the current timing a larger or smaller number than the specification? • If larger, advance the timing. • If smaller, retard the timing.	_____ _____ _____
5.	What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)?	_____
6.	Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates?	_____
7.	Use Table A to determine the current key part number. • What is the amount of the offset of the current key? • What is the part number of the current key?	_____ _____ _____

Table A

Timing Key Part Number	Offset	
	mm	in
216297	0.09	0.004
200711	0.18	0.007
3826781	0.23	0.009
216782	0.28	0.011
3826782	0.31	0.012
200709	0.38	0.015
3826783	0.43	0.017
200704	0.51	0.020
3826784	0.55	0.022
200708	0.54	0.021

Table A			
Timing Key Part Number	Offset		
	mm	in	
3000491	0.69	0.027	
200708	0.84	0.033	
3000491	0.91	0.038	
200714	0.99	0.039	
3000493	1.09	0.043	
3000494	1.10	0.47	
3000495	1.30	0.051	
8.	Use Table B to determine how to use Table C. Circle or check the appropriate answer.		
Table B			
Answer to Question 4	Answer to Question 6	Beginning Point on Table C	
Advance	Same	Top of column	_____
Advance	Opposite	Bottom of column	_____
Retard	Same	Bottom of column	_____
Retard	Opposite	Top of column	_____
9.	Answer the following questions BEFORE using Table C to determine the new timing key part number. - Find the current key part number listed at the top of the column on Table C. - Move up or down the column (the answer to question 8). Do not pass 0.000 (zero).		
If you pass 0.000 (zero), you will be choosing a key that does the opposite of what you want it to do.			
Stop when you locate the number nearest (± 0.002 in) to the required change in push rod travel (answer to question 5). Remain in this row. Move you finger to the right. The result is the new key part number and direction of offset the timing key must point.			
What is the part number of the new key?			

Note : Each column on Table C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The direction of offset will be pointing as indicated in the last column to the right on Table C.

Table C

S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	Part Number	Direction of Offset
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite
0.235	0.033	0.039	0.0345	0.0305	0.0268	0.042	0.0252	0.029	0.037	0.0415	0.045	0.0471	0.0491	3000494	Opposite
0.215	0.031	0.037	0.0325	0.0285	0.0248	0.040	0.0232	0.027	0.035	0.0395	0.043	0.0441	0.0471	3000493	Opposite
0.0185	0.028	0.034	0.0295	0.0255	0.0218	0.037	0.0202	0.024	0.032	0.0365	0.040	0.042	0.0441	200714	Opposite
0.018	0.0275	0.0335	0.029	0.025	0.0213	0.0365	0.0197	0.0235	0.0315	0.036	0.0395	0.0415	0.0436	3000492	Opposite
0.0155	0.025	0.031	0.0265	0.0225	0.0188	0.034	0.0172	0.021	0.029	0.0335	0.037	0.039	0.0411	200706	Opposite
0.0135	0.023	0.029	0.0245	0.0205	0.0168	0.032	0.0152	0.019	0.027	0.0315	0.035	0.037	0.0391	3000491	Opposite

Table C															
0.0 11	0.0 20 5	0.0 26 5	0.0 22	0.0 18	0.0 14 3	0.0 29 5	0.0 12 7	0.0 16 5	0.0 24 5	0.0 29	0.0 32 5	0.0 34 5	0.0 06 6	20 07 08	Op po sit e
0.0 09 5	0.0 19	0.0 25	0.0 20 5	0.0 16 5	0.0 12 8	0.0 28	0.0 11 2	0.0 15	0.0 23	0.0 27 5	0.0 31	0.0 33	0.0 35 1	20 07 04	Op po sit e
0.0 07	0.0 16 5	0.0 22 5	0.0 18	0.0 14	0.0 10 3	0.0 25 5	0.0 08 7	0.0 12 5	0.0 20 5	0.0 25	0.0 28 5	0.0 30 5	0.0 32 6	20 07 09	Op po sit e
0.0 05 5	0.0 15	0.0 21	0.0 16 5	0.0 12 5	0.0 09	0.0 24	0.0 07 2	0.0 11	0.0 19	0.0 23 5	0.0 27	0.0 29	0.0 31 1	21 67 82	Op po sit e
0.0 03 3	0.0 12 8	0.0 18 8	0.0 14 3	0.0 10 3	0.0 06	0.0 21 8	0.0 05	0.0 09	0.0 16 8	0.0 21 3	0.0 24 8	0.0 26 8	0.0 28 9	20 07 11	Op po sit e
0.0 01 7	0.0 11 2	0.0 17 2	0.0 12 7	0.0 08 7	0.0 05 1	0.0 20 2	0.0 03 4	0.0 02	0.0 15 2	0.0 19 7	0.0 23 2	0.0 25 2	0.0 27 3	21 62 94	Op po sit e
0.0 00	0.0 09 5	0.0 15 5	0.0 11	0.0 07	0.0 03	0.0 18 5	0.0 01 7	0.0 06	0.0 13 6	0.0 18	0.0 20 5	0.0 23 5	0.0 25 6	S- 30 2	N/ A
0.0 01 7	0.0 11 2	0.0 17 2	0.0 09 3	0.0 05 3	0.0 01 6	0.0 16 8	0.0 00	0.0 03 8	0.0 11 8	0.0 16 3	0.0 19 8	0.0 21 8	0.0 23 9	21 62 94	Sa me
0.0 03 3	0.0 06 2	0.0 12 2	0.0 07 7	0.0 03 7	0.0 0	0.0 01 52	0.0 01 6	0.0 02	0.0 10 2	0.0 14 7	0.0 18 2	0.0 20 2	0.0 22 3	20 07 11	Sa me
0.0 05 5	0.0 04	0.0 10	0.0 05 5	0.0 01 5	0.0 02	0.0 13	0.0 03 8	0.0 00	0.0 08	0.0 12 5	0.0 16	0.0 18	0.0 20 1	21 67 82	Sa me
0.0 07	0.0 02 5	0.0 08 5	0.0 04	0.0 0	0.0 03 7	0.0 11 5	0.0 05 3	0.0 01 5	0.0 06 5	0.0 11	0.0 14 5	0.0 16 5	0.0 18 6	20 07 09	Sa me
0.0 09 5	0.0 0	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me

Table C															
0.0 11	0.0 01 5	0.0 04 5	0.0 00	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Same
0.0 09 5	0.0 00	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Same
0.0 11	0.0 01 5	0.0 04 5	0.0 0	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Same
0.0 13 5	0.0 04	0.0 02	0.0 02 5	0.0 06 5	0.0 10 2	0.0 05	0.0 11 8	0.0 08	0.0 0	0.0 04 5	0.0 08	0.0 10	0.0 12 1	30 00 49 1	Same
0.0 15 5	0.0 06	0.0 0	0.0 04 5	0.0 08 5	0.01 2	0.0 03	0.0 13 8	0.0 10	0.0 02	0.0 02 5	0.0 06	0.0 08	0.0 10 1	20 07 06	Same
0.0 18	0.0 08 5	0.0 02 5	0.0 07	0.0 11	0.0 14 7	0.0 00 5	0.0 16 3	0.0 12 5	0.0 04 5	0.0 0	0.0 03 5	0.0 05 5	0.0 07 6	30 00 49 2	Same
0.0 18 5	0.0 09	0.0 03	0.0 07 5	0.0 11 5	0.0 15 2	0.0 0	0.0 16 8	0.0 13	0.0 05	0.0 00 5	0.0 03	0.0 05	0.0 07 1	20 07 14	Same
0.0 21 5	0.0 12	0.0 06	0.1 05	0.0 14 5	0.0 18 2	0.0 03	0.0 19 8	0.0 16	0.0 08	0.0 03 5	0.0 00	0.0 02	0.0 04 1	30 00 49 3	Same
0.0 23 5	0.0 14	0.0 08	0.0 12 5	0.0 16 5	0.2 02	0.0 05	0.0 21 8	0.0 18	0.0 10	0.0 05 5	0.0 02	0.0 00	0.0 02 1	30 00 49 4	Same
0.0 25 6	0.0 16 1	0.0 10	0.0 14 6	0.0 18 6	0.0 22 3	0.0 07 1	0.0 23 9	0.0 02 0	0.0 12 1	0.0 07 6	0.0 04 1	0.0 02 1	0.0 00	30 00 49 5	Same

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.

A working example is attached for your review to illustrate the use of this worksheet.

1.	What is the current timing?	0.267_____
2.	What is the timing code?	JF_____

3.	What is the timing specification for this code ($\pm$ 0.002 in)?	0.279_____
4.	Is the current timing a larger or smaller number than the specification? • If larger, advance the timing. • If smaller, retard the timing.	Smaller_____ Retard_____
5.	What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)?	0.012_____
6.	Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates?	Same_____
7.	Use Table A to determine the current key part number. • What is the amount of the offset of the current key? • What is the part number of the current key?	0.007_____ 200711_____

Table A

Timing Key Part Number	Offset	
	mm	in
216297	0.09	0.004
200711	0.18	0.007
3826781	0.23	0.009
216782	0.28	0.011
3826782	0.31	0.012
200709	0.38	0.015
3826783	0.43	0.017
200704	0.51	0.020
3826784	0.55	0.022
200708	0.54	0.021

Table A			
Timing Key Part Number	Offset		
	mm	in	
3000491	0.69	0.027	
200708	0.84	0.033	
3000491	0.91	0.038	
200714	0.99	0.039	
3000493	1.09	0.043	
3000494	1.10	0.47	
3000495	1.30	0.051	
8.	Use Table B to determine how to use Table C. Circle or check the appropriate answer.		
Table B			
Answer to Question 4	Answer to Question 6	Beginning Point on Table C	
Advance	Same	Top of column	_____
Advance	Opposite	Bottom of column	_____
Retard	Same	Bottom of column	_____X_____
Retard	Opposite	Top of column	_____
9.	Answer the following questions BEFORE using Table C to determine the new timing key part number. - Find the current key part number listed at the top of the column on Table C. - Move up or down the column (the answer to question 8). Do not pass 0.000 (zero).		
If you pass 0.000 (zero), you will be choosing a key that does the opposite of what you want it to do.			
Stop when you locate the number nearest ($\pm$ 0.002 in) to the required change in push rod travel (answer to question 5). Remain in this row. Move you finger to the right. The result is the new key part number and direction of offset the timing key must point.			
What is the part number of the new key? 200706			

Note : Each column on Table C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The direction of offset will be pointing as indicated in the last column to the right on Table C.

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